

1. Declaration

I, [Student Name], declare that this assignment, titled [Assignment Title], is my own original work and has not been copied from any other source except where explicitly acknowledged. I have not engaged in plagiarism, collusion, or any other form of academic misconduct in the preparation and submission of this assignment. All sources of information and data used in this assignment have been properly cited and referenced in accordance with the prescribed guidelines. I have not used unauthorized assistance in the preparation of this assignment and have not allowed any other student to copy my work. I am aware that any breach of academic integrity may result in disciplinary action as per the [policies of Monash University](#), which may include failing this assignment or the course, and further academic penalties.

Signature: _____Jui-Che Chen_____

Date: _____09/09/2025_____

2. Github Check

Enter your Github details here.

Github Username <i>Enter your username here</i>	Jche0548
Repository Shared? <i>Have you started and shared your assignment repository with your tutor yet?</i>	yes

3. Self-Evaluation

Rate your performance for each criteria. Put a ☒ (tick) in the box where you think your work belongs.

Criteria	Exceeds Expectations	Meets Expectations	Needs Improvement	Fail to meet expectations
BR (C.1): Authentication	☒			
BR (C.2): Role-based authentication	☒			
BR (C.3): Rating		☒		
BR (C.4): Security			☒	

4. Screen Recording of BRs

Create a 3 minute video showing your basic web application in action! Upload this video to your Google Drive and put the link here (ensuring that you have updated the access list so its not private).

<Link to Google Drive Video>

(make sure in the access settings you have shared it with your tutor OR set the permissions so that anyone with Monash account can video the video)

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5. Reflections: Implementation of C.4 Security

If you have implemented BR C.4, in less than 200 words describe the approach that you have taken to implementing Security in your application. What security flaws were you trying to prevent and what security measures have you implemented to fix those flaws? How do you know that these measures will help prevent those issues from happening? Optionally you can cite external sources to provide evidence for your claim.

I focused on preventing common client-side risks—especially XSS and tab-nabbing—using layered controls. First, all user inputs (e.g., Contact form) are validated and sanitized before storage: HTML tags and event handlers are stripped, emails are regex-checked, and fields have length limits. In the UI, I never use v-html; all user content is rendered via escaped Mustache bindings. I added a basic Content Security Policy (CSP) in index.html to block inline scripts and untrusted script origins. For external links opened in new tabs, I include rel="noopener noreferrer" to prevent tab-nabbing. I also avoid storing passwords in session state and wrap localStorage access in try/catch to fail safely.

I also verified the effectiveness through manual testing: injecting payloads like or <script>...</script> into the Message field does not execute; stored data appears sanitized in DevTools; and adding inline scripts triggers a CSP violation instead of running. Together, these measures reduce the XSS attack surface, mitigate link-based hijacking, and improve the overall robustness of client-side processing.

6. Reflections: Challenges

What has been the most challenging part of this assignment for you? How has this stretched you as a programmer?

For me, my biggest struggle with this project was combining login, registration, and role validation (admin vs user) into a single Home.vue page. Working with so many forms, error messages, ref, computed, and localStorage of user state and syncing state, made it easy to get stuff wrong. For example, the login message was not fading away and the login success message continued to appear after a user logged out! These were things that took a long time to debug.

Furthermore, I spent time understanding how Vue's lifecycle and navigation guards work with user state to restrict access to a page, for example only admin users could access the /admin page.

I also had some other challenges during the assignment, such as figuring out why success messages (i.e. "Login Successful") remained visible after logging out, debugging reactive logic in general, but especially when

navigating between App.vue, Home.vue, and auth.js. I needed to understand Vue's lifecycle hooks, two-way data binding, and the criticality of being consistent with naming and scoping (e.g. msg, or ok). Unfortunately, after all of this, the problem still exists. It may take me a bit longer to thoroughly investigate the underlying reasons behind the issue.

This assignment has truly empowered me to move beyond simply designing screens and move on to thinking about the entire application's state management, form validation, login logic, and front-end security (such as preventing XSS). Although there were many setbacks along the way, completing it gave me a real sense of accomplishment, and it has further advanced my understanding of Vue.

7. Declaration: Additional Help

Any tools that you used (including Gen AI or existing code reuse) must be declared here.

Note: GenAI is not allowed for coding purposes in any assignment,

However, you may use GenAI for brainstorming and problem solving. You need to declare all such uses here. One row per help used.

Name	Description
<i>Example: ChatGPT for brainstorming ideas</i>	<i>I used ChatGPT to brainstorm how to do X because I was feeling stuck with Y problem.</i>
ChatGPT for understanding how to modifying the authentication state	I used ChatGPT to understanding how to modifying the authentication state and follow the given step to modify.
ChatGPT for understanding Vue's lifecycle hooks, two-way data binding	The biggest issue I faced was why I could still see the success message (i.e. "Login Successful") after logging out, so I spent some time using ChatGPT to understand the theory behind two-way data binding and try to figure out what the main issue was.
ChatGPT for understanding Vue's lifecycle and navigation guards	When setting user status to restrict page access, I used ChatGPT to understand the Vue lifecycle, the theory and practice of navigation guards.
Ask ChatGPT how can I modify the red border warning in Contact page	According to the guidelines, cancel the is-invalid red border warning and use a small red text message to remind users instead.
ChatGPT to figure out why I was getting logged out	While inspecting the page, I found that whenever I refreshed the page, I would be forced to log out regardless of my role. I used ChatGPT to find the cause, and it turned out to be because I had cached the login status in two different keys, and the initial

when I refreshed the page.

fetches keys were different. After the correction, I used the same storage key to save and read the current user, and removed the non-existent save() and unused persist() methods.